

KST

Reproducing Glot500 on a Controlled 102-Language Setting with Vocabulary-Extension Initialization

Status: execution draft, refreshed 2026-06-28.

This draft is written as a paper-style report. Unresolved result slots must be filled only from measured artifacts in docs/exp/v5/3_evaluation/09_aggregation/ or from explicit blocker/status notes.

Live checkpoint progress and post-checkpoint Go/No-Go should be read from docs/exp/v5/4_reporting/final_action_dashboard_ko.md and bash scripts/run_v5_post_checkpoint_evals.sh status, not from this static draft header.

Abstract

We reproduce the core Glot500 training and evaluation pattern on a controlled 102-language-script setting composed of 92 XLM-R-seen Glot500 language-scripts and 10 diverse Glot500-internal target language-scripts. The experiment keeps the Glot500-style corpus merge, SentencePiece vocabulary expansion, continued

MLM pretraining, and downstream metric families, while intentionally reducing the language and compute scale. Our novelty is a focused comparison of initialization policies for newly appended vocabulary rows. In addition to the standard random resize baseline, we evaluate mean initialization and source-token decomposition initialization, which initializes a new token by averaging source embeddings obtained from the original tokenizer decomposition of that token surface. Before continued MLM, source-token decomposition substantially reduces target MLM proxy loss compared with random resize. Final after-MLM and downstream transfer claims remain conditional on matched v5_random and v5_fvt checkpoints and on available downstream task coverage.

1. Introduction

Multilingual language models such as XLM-R cover many languages, but their tokenization and representation quality remain uneven across language, script, and resource level. Glot500 addresses this limitation by scaling multilingual corpora and continued pretraining to more than 500 languages. Reproducing the full Glot500 experiment is expensive, but its experimental pattern can be studied in a controlled subset.

This report focuses on a 102-language-script replay. The head group consists of 92 local Glot500 language-scripts that are seen by XLM-R. The target group

consists of 10 language-scripts selected from Glot500 raw data that are not XLM-R seen and have at least 30,000 rows. This setting lets us ask two linked questions. First, can we faithfully execute the Glot500-style tokenizer, continued pretraining, and metric pipeline in a controlled setting? Second, when the tokenizer is expanded, does the initialization of newly appended embedding rows affect early adaptation quality?

The contribution is therefore not a new corpus. It is an experiment design that separates Glot500-style reproduction from a vocabulary-extension initialization novelty. The current strongest measured evidence is intrinsic: source-token decomposition initialization improves zero-step target MLM proxy loss over random and mean initialization while preserving source rows, special-token behavior, byte-row accounting, and LM-head tying.

1.1 Contributions

This report makes three bounded contributions.

First, it builds an artifact-audited controlled replay package for the Glot500-style pipeline over 92 XLM-R-seen language-scripts and 10 Glot500-internal target language-scripts. The package ties together corpus merge evidence, SentencePiece append-style tokenizer expansion, continued MLM handoff, metric-family coverage, and claim-freeze audits, while explicitly

rejecting a full-scale 511-language reproduction claim.

Second, it isolates vocabulary-row initialization as the method novelty. The source-token decomposition initializer preserves source row identity, remaps <mask> by token identity, accounts for byte and fallback rows, keeps the LM head tied, and lowers zero-step target weighted NLL relative to random resize.

Third, it keeps the Glot500 downstream metric families in the protocol rather than dropping coverage-limited tasks. PPPL, retrieval, classification, tagging, and Roundtrip alignment are all routed through aggregation, insertion, and claim-promotion gates; target10 downstream gaps remain visible limitations until measured task coverage exists.

2. Related Work And Positioning

Glott500 is the reproduction target for this experiment. The original Glott500 paper scales multilingual corpus construction and continued pretraining to more than 500 languages, and its evaluation surface motivates the v5 requirement to retain PPPL, retrieval, classification, tagging, and roundtrip-style metric families. In v5, this citation supports the experimental pattern, not a claim that the full 511-language training scale has been rerun. The local evidence for the subset replay is instead the 92+10 merge report, tokenizer audit, metric mapping, and coverage tables.

Yamaguchi et al. study low-resource continued vocabulary expansion and motivate the broader question of how newly introduced vocabulary rows should be handled.

In this report, Yamaguchi-style refers to that inspiration and to the earlier add-token route used in prior attempts. The main v5 tokenizer is deliberately different: it follows the Glot500-style SentencePiece append route. The novelty claim is therefore narrower and local to v5: after appending vocabulary rows, we compare initialization policies for those rows, with source-token decomposition as the main method candidate.

The citation boundary is maintained in:

`|docs/exp/v5/4_reporting/03_final_report/citation_source_map.md`

3. Scope And Reproduction Boundary

The allowed reproduction claim is:

`|We reproduce the Glot500-style training/evaluation pattern on a controlled 102-language subset.`

The disallowed claim is:

`|We fully reproduce the original 511-language Glot500 experiment.`

This distinction is important. The v5 run keeps the experimental logic of Glot500, including SentencePiece append-style vocabulary expansion, continued MLM pretraining, head/tail/all reporting, and the retained metric families.

However, it uses a smaller language set and a smaller compute budget.

cis-lmu/glot500-base is therefore an external reference model, not an equal-budget baseline.

The detailed fidelity matrix is maintained in:

`docs/exp/v5/4_reporting/00_tables/table_15_glot500_reproduction_fidelity.md`

The overview pipeline figure is:

`docs/exp/v5/4_reporting/01_figures/generated/figure_01_experiment_pipeline.png`

The short interpretation is that v5 is faithful at the level of experimental logic: Glot500-internal data, tokenizer expansion, continued MLM, required metric-family accounting, and head/tail/all-style reporting. It is intentionally not faithful at the level of full language scale or original compute budget.

4. Data

The v5 corpus contains 92 head language-scripts and 10 target language-scripts.

The target group is selected from Glot500 raw data with the following criteria:

not XLM-R seen, `new_length >= 30000`, local raw directory available, and regional/script diversity.

Target10:

`fur_Latn`
`krc_Cyrl`
`acm_Arab`
`dzo_Tibt`
`sat_Olck`
`mad_Latn`
`bam_Latn`

kjb_Latn
quw_Latn
rap_Latn

The full merge completed successfully.

Item	Value
head language-scripts	92
target language-scripts	10
source seen sentences	1,025,635,434
source target sentences	363,421
planned total samples	92,452,251
actual total samples	92,452,251
missing language dirs	0
merged file size	19G

Primary artifacts:

- corpus:

/home/axt/mnt2/jongha/v5_glot50010/data/
Glot500_v5_glot50010_xlmr100.txt

- manifest:

docs/exp/v5/0_tokenizer/merge/
Glot500_v5_glot50010_xlmr100.manifest.tsv

- report:

5. Tokenizer Expansion

The main tokenizer follows the Glot500-style SentencePiece append route. An

auxiliary SentencePiece tokenizer is trained on the controlled mixed corpus.

Pieces that are absent from the original XLM-R SentencePiece model are

appended to the base model.

This differs from the earlier Yamaguchi-style add-token route used in prior

experiments. The add-token route preserves existing tokenizer ids more

directly, while the Glot500-style append route is closer to the target

reproduction method but requires explicit special-token auditing. In v5,

<mask> moves from source id 250001 to target id 368686, so model rows

must be copied by token identity rather than by id prefix.

Tokenizer result:

Item	Value
base vocab size	250,002
extended vocab size	368,687
appended token count	118,685
byte fallback rows	256
source <mask> id	250001

Item	Value
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target <mask> id	368686
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audited languages improved	29/30
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target10 improved	9/10
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The main tokenizer reduced tokens per word for most audited languages. The

main caveat is dzo_Tibt, which worsened from 4.223938 to 5.552124

tokens/word. The regression is not explained by <unk> or byte fallback. It

appears to be a score-calibration/fertility issue where newly appended Tibetan

pieces can outcompete useful original pieces.

6. Embedding Initialization

We compare three initialized checkpoint families:

Method	Description	Role
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random	Hugging Face resize with random new rows	Glott500-style baseline
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mean	source/global mean initialization	simple stable ablation
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fvt	source-token decomposition mean	main novelty candidate
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For all non-random checkpoints, source rows are copied by token identity. The

source <mask> row is explicitly remapped to the target <mask> id. Byte rows

and fallback rows are counted separately, and input embeddings and LM head are

checked for consistency.

Main FVT initialization audit:

Audit item	Value
source identity rows copied	250,002
new token rows	118,685
FVT-initialized rows	118,427
byte/global-mean rows	256
global mean lexical fallback rows	2
<mask> max absolute diff	0.0
LM head tied	true

7. Continued MLM Pretraining

The main after-training comparison requires matched v5_random and v5_fvt checkpoints. The paired launcher trains v5_random_mlm_10k first and then v5_fvt_mlm_10k with the same corpus, tokenizer, schedule, seed family, and checkpoint interval.

Current live state is recorded in:

`docs/exp/v5/3_evaluation/running_status.md`

At the current check, v5_random_mlm_10k has produced the selected 10K checkpoint, while v5_fvt_mlm_10k is still running and has not yet produced a

model file. This live status is not a result claim, and post-checkpoint

evaluation remains locked until both rows report `ready_for_wrapper=yes` and

`post_checkpoint_preflight.md` reports

`post_checkpoint_preflight_ready_to_launch`.

Setting	Value
base model	xlm-roberta-base
tokenizer	v5 extended tokenizer
objective	MLM
max length	512
learning rate	5e-5
max steps	10,000
checkpoint interval	10,000
primary comparison	v5_random vs v5_fvt

8. Evaluation Setup

All Glot500 metric families are retained as required evaluation targets.

Coverage limitations are recorded rather than silently dropping a metric.

Metric	Score	Current coverage	Current status
PPPL / MLM proxy	weighted PPPL	102/102, target10 10/10	XLM-R, Glot500-base, and v5-random rows measured; FVT

Metric	Score	Current coverage	Current status
			waiting
Tatoeba retrieval	Top-10 accuracy	63/102, target10 0/10	XLM-R, Glot500-base, and v5-random rows measured; FVT waiting
Bible retrieval	Top-10 accuracy	74/102, target10 0/10	XLM-R, Glot500-base, and v5-random rows measured; FVT waiting
Text classification	macro-F1	1/102, target10 0/10	XLM-R, Glot500-base, and v5-random rows measured; FVT waiting
NER	F1	78/102 materialized coverage, target10 0/10; actual runner includes fur_Latn	XLM-R baseline, Glot500-base reference, and v5-random rows measured; FVT waiting
POS	F1	58/102, target10 0/10	XLM-R baseline, Glot500-base reference, and v5-random rows measured with TRAIN_LANGS=tur_Latn; FVT waiting
Roundtrip alignment	accuracy	74/102, target10 0/10	XLM-R, Glot500-base, and v5-random rows measured; FVT waiting

The target10 coverage boundary is central to interpretation. Target10 claims

should focus on tokenization, zero-step initialization diagnostics, and after-MLM PPPL unless target10 downstream task data is later materialized. Downstream results should be reported over available languages with explicit head/all and coverage notes.

For final assembly, the metric-by-metric fidelity boundary is also summarized in 00_tables/table_13_metric_fidelity_matrix.md. This table is the canonical bridge between the reproduction claim and the local execution state: each Glot500 metric family is either measured for the relevant model columns, waiting on matched v5 checkpoints, coverage-limited, or explicitly blocked.

8.1 Glot500 Fidelity Map

The experiment is designed as a faithful subset replay rather than a scale match. Each Glot500 component is retained where local data and inherited runners make it executable; otherwise the missing component is kept as an explicit blocked metric instead of being removed from the protocol.

Glot500 component	v5 implementation	Evidence	Status
multilingual corpus construction	92 XLM-R-seen Glot500 language-scripts + 10 Glot500-internal target language-scripts	merge report has 92,452,251 actual samples and 0 missing dirs	complete

Glott500 component	v5 implementation	Evidence	Status
SentencePiece vocabulary expansion	auxiliary SPM pieces appended to XLM-R SPM	extended vocab 368,687, appended rows 118,685	complete
continued MLM pretraining	paired v5_random then v5_fvt 10K run	launcher active on GPUs 2,3 in progress	
intrinsic PPPL / MLM proxy	wrapper and aggregation over 102 language-scripts	xlmr_base, glott500_base, partial, FVT row and v5_random rows measured	pending checkpoint
sentence retrieval	Tatoeba runner retained	xlmr_base, glott500_base, and v5_random rows measured over available languages	partial, FVT row pending checkpoint
text classification	Taxi1500 runner retained	xlmr_base, glott500_base, and v5_random rows measured on local English split	partial, coverage-limited, FVT row pending checkpoint
sequence tagging	NER and POS runners retained	NER xlmr_base/glott500_base/ v5_random measured; POS xlmr_base/glott500_base/ v5_random measured with TRAIN_LANGS=tur_Latn	partial, FVT rows pending checkpoint

Glott500 component	v5 implementation	Evidence	Status
Bible retrieval	metric family retained	74/102 local parallel data	
		materialized; XLM-R Top-10	partial, FVT row
		0.381153, Glott500-base Top-10 0.509356, v5-random Top-10 0.328019	pending checkpoint
roundtrip alignment	metric family retained	Bible-derived input set and v5 runner materialized; XLM-R accuracy 0.185300; Glott500-base accuracy 0.205189; v5-random accuracy 0.190300	partial, FVT row pending checkpoint

This map is the basis for final wording. It supports a claim of Glott500-style experimental fidelity on a controlled subset, but not a claim of full-scale Glott500 reproduction.

8.2 Result Promotion Rules

The report separates four evidence levels:

Evidence level	Can appear in final tables?	Example	Allowed wording
audited artifact	yes	merge report, tokenizer audit, init report, zero-step summary	proven setup or intrinsic result
parsed metric output	yes	non-empty test_results.txt	measured result

Evidence level	Can appear in final tables?	Example	Allowed wording
		or summary . tsv parsed by aggregation	
live run snapshot	no	current training step, dev F1 during tagging training	execution progress only
blocked or model-pending coverage row	yes, as limitation	FVT Bible/Roundtrip rows pending	retained metric with explicit status

This rule prevents three common overstatements. First, a live dev score is not a downstream result until the final prediction/evaluation file is complete and parsed. Second, `cis - lmu/glot500 - base` is an external reference, not an equal-budget baseline. Third, `target10` downstream improvement cannot be claimed while `target10` task coverage is absent outside PPPL.

The final conclusion should therefore be written in two layers. The first layer contains completed evidence: data scope, tokenizer expansion, initialization audits, zero-step FVT advantage, and measured baseline/reference rows. The second layer states the open test: whether the zero-step FVT advantage survives matched continued MLM and transfers to available downstream tasks.

9. Results

9.1 Tokenization

The tokenizer append run is structurally valid and improves audited fertility for 29/30 languages. It improves 9/10 target languages. The exception is dzo_Tibt, which remains a visible failure case.

9.2 Zero-Step Initialization

Before continued MLM, initialization differences are large on the v5 target group.

Method	Target weighted NLL	Head weighted NLL	Status
random	18.411756	12.895301	measured
mean	11.953142	8.037017	measured
fvt	8.785518	6.621457	measured

FVT improves target weighted NLL by -9.626238 compared with random resize and by -3.167624 compared with mean initialization. The central comparison artifact is docs/exp/v5/4_reporting/method_comparison_summary.md; in that summary, FVT reduces weighted NLL relative to random by 52.28% on v5 target, 48.65% on the head sample, and 51.31% on all measured zero-step rows. This supports the initialization novelty at zero step. It does not by itself prove after-MLM or downstream improvement.

9.3 Baseline And Reference Evaluation

The current parsed baseline/reference rows establish the comparison scale for later v5 method results.

Metric	Model	Group	Score	Value
PPPL	xlmr_base	v5 target	weighted PPPL	61.980216
PPPL	xlmr_base	head	weighted PPPL	8.117338
PPPL	xlmr_base	all	weighted PPPL	9.986271
PPPL	glot500_base	v5 target	weighted PPPL	15.102934
PPPL	glot500_base	head	weighted PPPL	10.213100
PPPL	glot500_base	all	weighted PPPL	10.640353
Tatoeba	xlmr_base	head	Top-10	0.656309
Tatoeba	xlmr_base	all available	Top-10	0.566067
Tatoeba	glot500_base	head	Top-10	0.743755
Tatoeba	glot500_base	all available	Top-10	0.706649
Tatoeba	v5_random	head	Top-10	0.700285
Tatoeba	v5_random	all available	Top-10	0.610353
Bible	xlmr_base	head/all available	Top-10	0.381153
Bible	glot500_base	head/all available	Top-10	0.509356
Bible	v5_random	head/all available	Top-10	0.328019

Metric	Model	Group	Score	Value
Taxi1500	xlmr_base	head/all available	macro-F1	0.592876
Taxi1500	glot500_base	head/all available	macro-F1	0.743338
Taxi1500	v5_random	head/all available	macro-F1	0.702956
NER	xlmr_base	all available	F1	0.549858
NER	xlmr_base	head	F1	0.621207
NER	xlmr_base	v5-target actual intersection	F1	0.459364
NER	glot500_base	all available	F1	0.627108
NER	glot500_base	head	F1	0.645915
NER	glot500_base	v5-target actual intersection	F1	0.553191
NER	v5_random	all available	F1	0.544628
NER	v5_random	head	F1	0.608020
NER	v5_random	v5-target actual intersection	F1	0.560554
POS	xlmr_base	all available	F1	0.481336
POS	xlmr_base	head	F1	0.571446
POS	glot500_base	all available	F1	0.567542
POS	glot500_base	head	F1	0.573832
POS	v5_random	all available	F1	0.481102

Metric	Model	Group	Score	Value
POS	v5_random	head	F1	0.587430
Roundtrip	xlmr_base	head/all available	accuracy	0.185300
Roundtrip	glot500_base	head/all available	accuracy	0.205189
Roundtrip	v5_random	head/all available	accuracy	0.190300

The NER v5-target row should be interpreted narrowly: it comes from the actual evaluated intersection and currently consists of fur_Latn only. It is not a target10-wide downstream result.

The POS rows are available-language tagging references. XLM-R-base, Glot500-base, and v5-random POS runs use TRAIN_LANGS=tur_Latn because the local POS materialization does not include train-eng_Latn.tsv. The measured v5-random POS row has F1 0.481102 over all available POS languages and 0.587430 over the head group; the matched FVT POS row remains checkpoint-gated.

9.4 Pending Main Method Results

The v5_random after-MLM PPPL row is now measured. Its weighted PPPL is 39.222875 on v5 target10, 18.726452 on the head group, and 20.138927 on all 102 language-scripts. This is a measured random-initialized checkpoint row, not yet a method comparison: the corresponding v5_fvt PPPL row is still

locked until the matched FVT checkpoint is ready and parsed by aggregation.

The v5_random Tatoeba retrieval row is measured with Top-10 accuracy

0.700285 on the 63-language head split and 0.610353 over all 98 available language-scripts. This fills the random-checkpoint available-language row; it is not a target10 downstream result because Tatoeba target10 coverage remains 0/10.

The v5_random Taxi1500 row is measured with macro-F1 0.702956 and accuracy 0.747748 on the local English-only split. This is a limited available-language classification row; it does not unlock a target10 downstream claim.

The v5_random Roundtrip row is also measured with accuracy 0.190300 over the same 74 available language-scripts as the XLM-R and Glot500-base reference rows. This is available-language downstream evidence for the random checkpoint only; downstream method claims remain locked until the matched v5_fvt Roundtrip row exists.

The v5_random Bible retrieval row is measured as well, with Top-10 accuracy 0.328019 over the same 74 available Bible language-scripts. This is a single-checkpoint measured row, not evidence that the embedding initialization method wins; that comparison still requires the matched v5_fvt Bible row.

The v5_random NER row is measured too, with F1 0.544628 over all 164

actual runner language rows, 0.608020 over the head group, and 0.560554 on the one-language v5-target actual intersection (fur_Latn). This is available-language tagging evidence for the random checkpoint only; it does not unlock a target10-wide downstream claim.

The v5_random POS row is also measured with F1 0.481102 over all available POS languages and 0.587430 over the head group under the same TRAIN_LANGS=tur_Latn condition as the baseline/reference POS rows. It is a random-checkpoint tagging row, not yet a method win; that comparison requires the matched v5_fvt POS row.

Taken together, the measured v5_random rows are best interpreted as a diagnostic lower-bound checkpoint. The random-initialized model improves target10 PPPL relative to XLM-R (39.222875 versus 61.980216) but is still substantially behind Glot500-base (15.102934), and its head/all PPPL remains worse than both references. Available-language downstream behavior is mixed: it improves over XLM-R on Tatoeba and Taxi1500 but not to the Glot500-base reference level; it is below both references on Bible retrieval; it is near XLM-R on all-language POS, above both references on head POS, and between XLM-R and Glot500-base on Roundtrip. This mixed profile strengthens the need for the matched v5_fvt test rather than weakening the design: the final

method claim must depend on whether source-token-decomposition initialization improves over this random-checkpoint baseline after the same 10K MLM budget.

The following rows are intentionally pending:

- v5_fvt after-MLM PPPL.
- v5_fvt Tatoeba retrieval.
- v5_fvt Bible retrieval over available Bible languages.
- v5_fvt text classification after the matched checkpoint.
- v5_fvt NER and v5_fvt POS over available task languages.
- v5_fvt Roundtrip alignment over available Roundtrip languages.

The required post-checkpoint insertion path is:

```
bash scripts/run_v5_post_checkpoint_evals.sh status
SKIP_MEASURED=1 WITH_PLOTS=1 GPU_RANDOM=0 GPU_FVT=1 bash
scripts/run_v5_post_checkpoint_evals.sh all
python3 scripts/refresh_v5_reporting.py --with-plots
```

Only rows parsed into 3_evaluation/09_aggregation/ should replace the pending labels above. If a metric remains unavailable, it must stay as

blocked-data or coverage-limited rather than disappearing from the results.

The exact launch environment, output/log paths, and promotion rule are generated in 3_evaluation/post_checkpoint_execution_plan.md.

9.5 Outcome-Conditioned Final Wording

After post-checkpoint evaluation, the abstract and conclusion should be

updated from the decision tree rather than rewritten ad hoc. The outcome matrix

for the Korean report/PPT is:

`docs/exp/v5/4_reporting/post_checkpoint_outcome_matrix_ko.md`

The English prose blocks are maintained in:

`docs/exp/v5/4_reporting/03_final_report/result_interpretation_blocks.md`

The file-by-file report/PPT edit order is maintained in:

`docs/exp/v5/4_reporting/post_result_patch_plan_ko.md`

The final report should choose exactly one of these outcome families:

Outcome family	Required evidence	Final wording rule
bounded positive	FVT wins after-MLM PPPL and most available downstream rows	claim a bounded method benefit, while keeping target10 downstream coverage caveats
intrinsic positive, downstream mixed	FVT wins after-MLM PPPL but downstream rows vary only; describe downstream as task- and by task	claim intrinsic adaptation improvement coverage-dependent
early-only diagnostic	FVT wins zero-step but not after matched MLM	frame initialization as early-adaptation evidence rather than final superiority
negative final comparison	FVT does not beat random in PPPL or downstream	report a controlled negative result and discuss training/objective implications
incomplete evaluation	matched rows or parsers remain missing	keep the execution-draft conclusion and leave final method claims locked

No final abstract or conclusion sentence should be upgraded unless

`post_result_patch_plan_ko.md`, `final_claim_decision_tree.md`,

`result_promotion_readiness_audit.md`, and `final_claim_freeze_audit.md`

agree on the same non-pending outcome and patch scope.

10. Analysis

The zero-step results suggest that new vocabulary rows are not a neutral implementation detail. Random resize creates a large target-group penalty before training. Mean initialization stabilizes the rows, but source-token decomposition produces the best current intrinsic result. The method appears especially useful when a new surface form can be decomposed into informative source subpieces by the original tokenizer.

The tokenizer result is also instructive. A larger vocabulary can reduce fertility for most languages but still harm an individual script when appended piece scores alter segmentation behavior. The dzo_Tibt case should remain in the main analysis because it prevents overclaiming and points to a concrete repair direction: score calibration or script-aware filtering of appended pieces.

11. Limitations

This is not a 511-language Glot500 reproduction. It is a controlled 102-language replay. The selected target10 has raw-text coverage but little downstream task coverage in the currently materialized evaluation data. Therefore target10 downstream improvement is not a supported claim.

Bible retrieval and roundtrip alignment are retained as required metric families. Bible retrieval now has available-language local parallel data materialized for 74/102 language-scripts, with target10 coverage still 0/10. XLM-R-base, Glot500-base, and v5-random rows are measured, while the remaining gap is the matched v5-FVT row after checkpoint completion and post-checkpoint preflight. Roundtrip alignment now has Bible-derived inputs, a v5 runner, and measured XLM-R-base/Glot500-base/v5-random rows over 74 language-scripts; its remaining gap is also v5-FVT execution after the matched checkpoint. These states should appear in final tables explicitly rather than being omitted.

cis-lmu/glot500-base is an external reference model with a different training scale. It is useful context, but it is not an equal-compute baseline.

Reproducibility Appendix

The full command/path appendix for reproducing or auditing v5 is maintained in:

`docs/exp/v5/4_reporting/03_final_report/reproducibility_appendix.md`

The appendix records the 92+10 scope lock, canonical local paths, merge and tokenizer commands, paired MLM launch, metric-wrapper pattern, model keys, required metric handling, result-promotion rules, and the post-result patch plan that keeps report/PPT edits tied to parsed aggregation rows.

12. Conclusion

The v5 setup establishes a faithful controlled replay of the Glot500-style workflow on a 102-language subset. The data scope is frozen, the full corpus merge is complete, the SentencePiece append tokenizer is trained and audited, and all Glot500 metric families are retained with coverage or blocker accounting. This is not a full 511-language reproduction, but it preserves the experimental structure needed to study tokenizer expansion, continued MLM, and head/tail/all evaluation under a tractable budget.

The completed method evidence already supports the central novelty motivation. Newly appended vocabulary rows are not an implementation detail: source-token decomposition initialization lowers v5-target zero-step weighted NLL from 18.411756 under random resize to 8.785518, while preserving source rows, remapping <mask> correctly, separating byte fallback rows, and keeping the LM head tied. The tokenizer audit is also mostly positive, improving fertility for 29/30 audited languages, with dzo_Tibt kept as a visible regression and analysis case.

The remaining conclusion depends on matched v5_random and v5_fvt continued-MLM checkpoints and the available downstream replay. Until those rows are parsed, the final claim should remain bounded: v5 demonstrates a controlled

Glott500-style replay and a strong intrinsic initialization advantage before training; whether that advantage survives MLM and transfers to downstream tasks is the next measured gate.

References

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